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SPORTS TRAINING APPARATUS AND METHOD

Abstract

An athletic training assembly includes a vest to be worn by an athlete. The vest includes attachment points at plural locations on both front and rear regions of the vest. A training coupler has opposite first and second ends, with an attachment mechanism affixed to at least the first end. The attachment mechanism is configured to be attached releasably to one of the attachment points of the vest. A region of the training coupler at or near the second end is configured to be held by a coach or affixed to a support. The training coupler is aligned to require an athlete to exert a force against the training coupler while performing an athletic movement, thereby developing muscles for exerting top-down forces when performing a specified athletic movement.

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Background/Summary

BACKGROUND

[0001] 1. Field of the Invention. The invention relates to an apparatus and method for training athletes, such as baseball and softball players, with respect to proper positioning of the upper body relative to the lower body and proper application of force from the upper body when performing various athletic maneuvers, such as baseball batting or baseball pitching. [0002] 2. Related Art. Coaches and trainers typically teach their student athletes and trainees that the forces applied when performing an athletic maneuver should be generated from the ground up. This approach might be compared to a spring that is uncoiling from a compressed condition adjacent to the ground. The ground-up theory has been applied to many sports, including: baseball, boxing, hockey, lacrosse, wrestling and track/field events.

[0003] Baseball batting is one of the most difficult maneuvers in all of sports. A successful batter is required to use a generally cylindrical bat to hit a generally spherical ball that is being thrown toward a strike zone without the batter knowing the location in the strike zone to which the ball will be pitched or the speed at which the ball will be pitched.

[0004] Many devices have been developed to help train batters for the difficult task of hitting a baseball. For example, batting tees are used in virtually all organized baseball leagues and at many homes. The typical batting tee includes a height-adjustable tube that extends up from a horizontal support that often resembles home plate. The upper end of the tube is dimensioned to support a baseball or softball. The batter will stand facing the tee and will repeatedly hit balls placed on the upper end of the tubular support. A coach may be available to help the batter achieve proper positioning of the hands, arms and shoulders prior to the start of the swing and to have legs and feet properly positioned at the start of the swing. A coach also can give advice as to proper movement of the leading foot at the start of the swing and proper movement of the hands and torso during the course of the swing. However, a coach generally cannot provide significant guidance while a swing is being completed because such hands-on guidance would interfere with the swing of the bat by the player who is trying to hit the ball and the movement of the bat can be dangerous to the coach. Furthermore, the stationary ball supported on a batting tee does not train the batter to develop appropriate movements and body alignments during the course of an attempt to hit a rapidly moving baseball or softball. Most importantly, training devices comparable to a batting tee do not help the trainee to learn how the rotational rates of force should be developed from the upper body during the course of a swing. Many such prior art approaches focus on speed with little or no emphasis on force and power. To the contrary, many such prior art approaches teach that attempts to increase force and power will detract from the desirable mechanics of arm and hand movements. [0005] Training devices for a batter also include a baseball that is suspended from a tether that has an upper end supported at a position above home plate. A tethered baseball avoids the need to reposition the ball on a tee after each swing and avoids the need to retrieve hit baseballs. However, hitting a tethered baseball is no more realistic of a game situation than hitting a ball from a tee. Additionally, a tethered baseball does not enable a coach to assist the batter to develop appropriate skills with respect to the position and alignment of the upper body during the course of a swing, proper application of force from the upper body or the rotational rates of force developed by the upper body and delivered during the course of a swing.

[0006] The prior art also includes rigid structures disposed in proximity to a batter for guiding the

movement of a bat during the course of swing. These devices do not simulate game conditions, do not help develop proper positioning or movement of the upper body relative to the lower body during the course of a swing and do not train the athlete to deliver increased force and power.

[0007] Proper positioning of the upper body during the course of pitching a baseball also is important and is difficult to train using known methods and devices.

[0008] The prior art also includes mat-like templates that can be positioned on the ground in a batter's box or on a pitching mound. These templates include indicia to identify where a batter's or pitcher's feet should be at the start of a movement and after completing a follow-through. However, these templates are not helpful for assisting the batter or pitcher to achieve proper positioning of the upper body during the course of a batting or pitching movement. Templates on the ground also are not useful for learning proper application of force from the upper body or the rotational rates of force developed by the upper body and delivered during the athletic maneuver to which the template pertains.

[0009] U.S. Pat. No. 7,572,198 issued to one of the inventors herein and relates to a baseball training bat to help batters achieve a proper grip of a bat during the course of a swing. The baseball training bat disclosed in U.S. Pat. No. 7,572,198 has been very effective to help develop proper hand positioning on a bat, but is not helpful for training baseball players for proper positioning and alignment of the upper body relative to the lower body during the course of the hitting or pitching maneuver and does not help to train the athlete for applying greater force during critical phases of an athletic maneuver.

[0010] In view of the above, an object of the invention is to provide an apparatus that can be useful for training athletes to develop skills to achieve proper alignment of the upper body relative to the lower body during the course of performing an athletic maneuver and to develop a more effective rotational force and acceleration from the upper body during the course of a baseball swing, a pitching maneuver or many athletic maneuvers in other sports, such as: golf, tennis, boxing, wrestling, lacrosse, hockey, shot putting and virtually all other field events.

SUMMARY

[0011] The apparatus and method disclosed herein takes a fundamentally different approach than the prior art described above. In this regard, the apparatus and methods described herein enable an athlete to train in a way that generates more force from the upper body in a generally top to down direction, rather than in the opposite direction. Thus, the apparatus trains the athlete to create force from the top down. The upper body is pinning the legs down into the ground like the spring of a pogo stick. The legs get pinned between the upper body and the ground, resulting in faster rotational speeds of the torso. This is like creating stored energy by compressing a spring and then releasing the stored energy as the spring uncoils, thereby converting that stored energy into kinetic energy. The apparatus and methods described herein enhance rotational rates of force from the upper body and help to develop muscle groups that will apply force in this manner. Many aspects of this disclosure focus on baseball, but the apparatus and methods described in this disclosure relate to virtually all athletic activities where strength and speed are an asset.

[0012] One aspect of this disclosure relates to a training vest that can be worn by an athlete, such as a baseball player, to help train the athlete regarding proper positioning and alignment of the upper part of the athlete's body relative to the lower part of the athlete's body while performing an athletic maneuver and to achieve a very effective linear force or rotational force by and from the upper part of the body relative to the lower part of the body. The term vest as used herein includes a closely fitting garment that surrounds at least part of the torso of the athlete while permitting at least a pivoting movement of the torso and free movement of at least one arm. In some embodiments, the vest may resemble a corset. In other embodiments the vest may resemble an upper part of a wetsuit worn by a surfer, provided that such a garment is dimensionally stable so as not to stretch in response to pulling or twisting forces. Unlike a wetsuit, the vest of this disclosure leaves the arms free to move independent of the vest and the torso. In some embodiments, the lower end of

the vest is above the waist so as not to impede pivotal movement of the torso relative to the pelvic region of the athlete. The pivoting axis of the torso may extend along or parallel to the spine or in a diagonal direction. The vertical dimension of the vest preferably is uniform around the entire vest. Thus, the vest can be reversed front-to-rear depending on whether the vest is to be tightened by the coach or the athlete. A coach can tighten the vest from the rear, and the athlete can tighten the vest more easily from the front when the athlete is working independently from the coach.

[0013] The vest of some embodiments includes a front panel to be positioned on the chest of the athlete, a rear panel to be positioned on the back of the athlete and first and second shoulder straps that connect the front and rear panels while being positioned and supported on the respective shoulders of the athlete.

[0014] The vest of some embodiments further includes connecting panels that extend from at least one of the front and rear panels to wrap around the sides of the athlete. Hook and loop fasteners, such as VELCRO® fasteners, can be provided on the connecting panels. For example, a fabric with an array of loops can be positioned on the surface region of each connecting panel remote from the front or rear panel from which the connecting panel extends. The connecting belts can be provided with a hooked fabric that can releasably engage the looped fabric on the front panel or the rear panel. Strap locking mechanisms can be provided instead of hook and loop fasteners. Such strap locking mechanisms may be similar to those used for securing loads to roof racks or similar to those used in infant car seats.

[0015] The athlete wears the vest of this embodiment by positioning the first and second shoulder straps on the respective shoulders so that the front panel lies on the chest of the athlete and the rear panel lies on the back of the athlete. The athlete or the coach then can position the connecting panels adjacent the respective sides of the athlete. The hook fasteners then can be secured to the mating loop fasteners to keep the vest held snugly on the athlete. The vest of some embodiments is formed from a flexible dimensionally stable fabric or a flexible dimensionally sheet material so as to conform to the body of the athlete without restricting pivoting movement of the torso.

[0016] The vest further includes at least one attachment point, such as a ring, on at least the rear panel. In some embodiments, at least one ring is provided on the front panel. Plural rings can be provided on the front panel and/or on the rear panel. In some embodiments, plural rings are at laterally spaced positions in proximity to an upper edge of the rear panel and plural rings can be provided at laterally spaced positions along an upper edge of the front panel. Similarly, at least one ring can be provided in proximity to the lower edge of the rear panel and/or along the lower edge of the front panel. One or more rings also can be provided on or near the shoulder straps.

[0017] The vest is used with at least one elongate training cord or training coupler having opposite first and second ends. The coupler may be dimensionally stable cord or strap, but in most embodiments described herein the coupler exhibits a specified and desired elasticity. An elastic training coupler that exhibits 50-80 pounds resistance is suitable for most instances. The term coupler is intended to define a connection that can extend between the athlete and either a coach or a support. The training coupler of most embodiments is flexible and typically has a length of 3-8 feet. A first hook, clip or other fastening means may be provided on the first end of the training coupler and may be configured to be connected releasably to at least one of the rings on the training vest. The second end of the coupler of some embodiments may have a second hook or clip that can be held by a coach or can be secured to hook or ring on a rigid support. Additionally, the first and second hooks or clips can be attached to two different rings or attachment points on the vest, and the coach can hold an intermediate position to easily vary the forces applied at the two positions during the course of the athletic maneuver being performed by the athlete.

[0018] The vest or other such garment can be used, for example during a baseball batting or pitching practice. The vest that is held snugly against the chest and back of the batter does not interfere with a batting or pitching maneuver. In some training exercises, the hook or the clip on the first end of the coupler can be releasably connected to one of the rings along the top edge of the

rear panel while the second end of the coupler is held by the coach. Neither the coupler nor the coach interfere with the movement of the torso and the arms during batting or pitching and similarly do not interfere with the bat during a batting maneuver. However, the coach can maintain or apply appropriate tension on the second end of the coupler for maintaining proper alignment of the upper body relative to the lower body during a batting or pitching maneuver and, in some instances, to urge the athlete toward an unstable position. The athlete will urge his or her body counter to the force exerted by the coupler in an effort to regain stability, and these forces exerted by the athlete to counter the forces applied by the coupler will vary the performance of the athletic maneuver in a way that will increase the power applied during the performance of the athletic maneuver. In some instances, the tension on the coupler will enable the batter to lean slightly forward while in a batting stance and waiting for the pitch. However, the coach can apply additional tension either to prevent the batter from lunging into the pitch or to encourage the batter to lunge into the pitch and to prevent or allow shifting weight rearwardly during the swing. The batter or pitcher, on the other hand, can practice by applying increased force opposite to the force applied by the coupler.

[0019] The coach can select the particular ring to which the hook on the training cord or coupler will be connected in accordance with the emphasis that must be used for a particular batter or pitcher based on prior observations by the coach. Most athletic maneuvers require pivoting at least a part of the body of the athlete about a pivot axis, and the pivot axis often will extend along the spine of the athlete. The connection between the training cord or coupler and the ring on the vest often will be at a distance spaced radially outward from the pivot axis for the particular athletic maneuver, such as the swinging of the baseball bat or the throwing or pitching of a baseball. To complete the athletic movement, the athlete will be trained and/or forced to overcome the force applied by the coupler to the part of the body that is completing the athletic movement. The combination of the vest and the training coupler enable the coach, in some instances, to move forward in the direction of the athlete (e.g. the pitcher) or rearward away from the athlete (e.g. the pitcher) depending upon the direction of the tension force that is required for a particular batter. Additionally, a training exercise can have the batter repeat swings while moving successively toward the pitcher, while the coach also is moving forward toward the pitcher but at a position behind the batter and with little or no direct effect on the batter other than the effect achieved by the forces applied by the coupler to the vest and hence to the athlete to which the vest is snugly fit. Furthermore, with each of these successive swings, the coach ensures that the connection between the coupler and the ring on the vest defines the best radial position from the pivot axis (e.g. the spine) for the rotation about the pivot axis that develops the appropriate directional force that is applied by the upper body during the performance of the athletic movement.

[0020] The training coupler need not extend linearly from the first end to the second end. For example, the first end of the training coupler can be connected to a ring on the front panel. The training coupler may then extend around the side of the athlete and may be held by a coach facing the back of the athlete. Thus, the training coupler may undergo a more than 90° change of direction between the first and second ends, and in some instances at least a 135° change in direction. A pulling force exerted on the second end of the training coupler by the coach will be converted into a rotational force on the torso of the athlete, in much the same manner as a pulling force exerted on a starter cord of a chainsaw or snowblower. The athlete will exert forces counter to the forces applied by the training coupler, thereby developing muscle groups and muscle memory for producing powerful torque.

[0021] Plural training couplers can be used simultaneously. For example, an upper training coupler can be coupled to the upper right ring on the rear panel of a vest worn by a right handed batter while a lower training coupler can be coupled to the lower right ring on the rear panel of the vest worn by that batter. A coach then can apply tension forces simultaneously to the second ends of the upper and lower training couplers. The forces applied to the upper and lower couplers can be

essentially equal throughout the swing, or different forces can be applied at various points in the swing to achieve the effect intended by the coach. In other situations, the first ends of the upper and lower training couplers can be connected to upper and lower rings on the front panel of the vest. The upper and lower training couplers then can wrap around the side of the athlete and the second ends of the upper and lower training couplers can be held by the coach at a position rearward of the athlete. As explained above, the training couplers used in this manner undergo more than 90° changes of direction, and in some instances at least a 135° change in direction. Pulling forces exerted by the coach on the second ends of the upper and lower training couplers will be converted into rotational forces on the torso of the athlete. The athlete will exert rotational forces counter to the forces applied by the upper and lower training couplers, and those counteracting forces exerted by the athlete help the athlete to develop muscle groups and muscle memory for producing powerful torque.

[0022] The vest also can be used for training purposes other than batting. For example, the vest can be worn by a pitcher while the coach is applying appropriate tension to the training coupler for controlling the alignment and positioning of the upper part of the pitcher's body during the course of a pitching maneuver and for defining the pivot point or pivot axis from which upper body forces are developed.

[0023] Some training will have the coach positioned in front of the athlete and with the hook on the first end of one end of one or more training couplers connected to one of the rings on the front panel of the training vest. This use of the training couplers can be helpful for wrestlers who often exert rearward driving forces by their upper backs and shoulders against an opponent followed by rotation into a position facing an opponent.

[0024] The angle of the training coupler will be chosen based on the desired movement of the torso during performance of the athletic maneuver. For example, the second end of the training coupler may be higher than the first end for training exercises for training pitchers. For example, instability applied to the body from above its center of gravity forces the human body to find stability down below. A puppeteer can only control the puppet from the top, not from the bottom. This same principle can be used with the subject invention by applying forces to the vest while the second end of each training coupler is positioned higher than the vest. On the other hand, some athletic maneuvers can benefit from having the second end of the training coupler lower than the first end.

[0025] Many athletic training devices are considered to be intended for younger athletes who are learning a sport. However, the vest and training coupler described herein also have been found to be particularly effective for top tier athletes who are striving for perfection. For example, the vest and training coupler described herein has been used with a professional pitcher who had a top pitching speed of approximately 93 mph. Use of the training vest and training coupler for several days enabled the pitcher to reach speeds of 98 mph. A 5 mph increase in the pitching speed of a young developing athlete is not surprising, but a 5 mph increase in maximum pitching speed by a professional baseball pitcher is very significant.

[0026] The preceding descriptions have focused on baseball training. However, the training vest and the training cord or coupler have application in other sports, as described above, such as boxing, tennis, golf, hockey and lacrosse where force can be applied by a pivoting movement by an upper part of the body, such as a pivoting movement about the spine. More particularly, the combination of the training vest and the training coupler can be used with at least the following athletic movements: an aerobic exercise, an archery, dart or shooting motion, a baseball or softball swing or pitch, a basketball dribble, pass or shot, a billiards shot, a bowling motion, a cheerleading motion, a climbing motion, a crawling motion, a cricket swing, a cycling motion, a dance motion, a football throw or kick, a golf swing, a gymnastics motion, a hockey swing, a lacrosse motion, a polo swing, a punch, kick or throw, a skateboard or surfing motion, a skating motion, a skiing motion, a soccer, rugby or football kick, a tennis, badminton, picket ball or handball swing, a track and field motion, a volleyball bump, set or serve, a walking, jogging or running motion, a water

bag exercise or a weightlifting exercise. Thus, the combination of the training vest and training coupler can be used so that the force exerted by the training coupler comprises at least one of a linear force, a rotational force or an oppositional force. The particular attachment point of the first end of the training coupler to the vest and the alignment of the training coupler extending from the vest can enable the athlete to develop or emphasize muscles for delivering a top-down force, a stronger pivoting force about a pivot axis, such as the spine, and to expose disadvantageous movement patterns by accentuating the role of such disadvantageous movement patterns in an athletic maneuver and/or to elicit a counter response by the athlete

[0027] The preceding descriptions have focused primarily upon the training vest worn by the athlete and the training cord or coupler held by a coach. However, the end of the training cord or coupler remote from the vest can be affixed to a stationary support on a wall, floor, ceiling or post. A vertical support for affixing the second end of the training coupler can be configured to provide plural vertically spaced attachment points of the second end of the training coupler to the vertical support, thereby enhancing and developing the proper alignment of the top-down application of force by the athlete when performing the athletic maneuver.

[0028] The combination of the training vest and the training coupler also can be supplemented with electrical, electromechanical and electro-optical devices. For example speed sensors, force sensors or accelerometers can be applied to or incorporated into the training coupler to measure forces, speed and/or acceleration developed during the course of an athletic maneuver. Signals produced by these sensors can be transmitted wirelessly or by wired connection to a control unit that measures, records and analyzes various force, speed and acceleration levels over time for subsequent analysis or for effecting subsequent changes in force levels applied by the training cord or coupler.

[0029] A “control unit” can be understood in connection with the invention to mean a machine, processor or an electronic circuit. In particular, a processor can be a main processor (central processing unit (CPU)), a microprocessor, or a microcontroller, for example an application-specific integrated circuit or a digital signal processor, possibly in combination with a memory unit for storing program instructions, etc. A processor can also be understood to mean a virtualized processor, a virtual machine, or a soft CPU. For example, the processor can also be a programmable processor equipped with configuration steps for carrying out the above-mentioned training method according to the invention or configured with configuration steps in such a way that the programmable processor contributes to achieving the features and objects of the invention that could not be appreciated by the naked eye. In particular, the processor can include highly parallelized computing units and high-performance graphics modules and monitors.

[0030] In connection with the invention, a memory unit or storage module can be a volatile memory in the form of a working memory (random access memory, RAM), or a permanent memory such as a hard drive or a data carrier or, for example, a replaceable memory module. However, the storage module can also be a cloud-based storage solution. An interface also may communicate with the control unit for inputting information to the control unit and for outputting information. The interface may comprise a touch screen or keyboard for inputting information or command and a monitor for displaying results, such as performance data by the athlete.

[0031] The training vest, the training coupler also can be used with video devices that may communicate with a version of the above-described control unit. For example, force units as measured by the force measuring devices can be used so that force levels can be correlated to the video data that defines the stage of the athletic movement. These data can be transmitted to a part of the training coupler for increasing or decreasing forces applied. The video data also can track movement paths of particular rings on the vest or movement paths of indicia on the vest, such as indicia on or along the pivot axis (e.g. the spine) or at specified distances spaced radially from the pivot axis.

[0032] These and other aspects of the invention will become more apparent in the following description of a specific embodiment with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] FIG. 1 is a top plan view of the vest in a flattened disposition and showing the surfaces of the vest that will face outwardly when the vest is in use.

[0034] FIG. 2 is a bottom plan view of the vest in the flattened disposition and showing the surfaces of the vest that will face toward the person wearing the vest when the vest is in use.

[0035] FIG. 3 is a front view of a player wearing the vest.

[0036] FIG. 4 is a rear view of the player wearing the vest.

[0037] FIG. 5 is a rear view of a right-handed batter wearing the vest having the training cord attached thereto.

DETAILED DESCRIPTION

[0038] A training vest in accordance with an embodiment of the invention is identified generally by the numeral **10** in FIGS. 1-5. The training vest **10** includes a rear panel **30** and a front panel **40** to be disposed respectively adjacent the back and the chest of an athlete. The rear panel **30** and the front panel **40** are generally rectangular in the illustrated embodiment. However, other configurations of the rear and front panels **30** and **40** are possible. The rear panel **30** has opposite top and bottom edges **32** and **34** respectively and opposite first and second side edges **35** and **36** respectively. Additionally, the rear panel **30** has an outer surface **37**, as illustrated in FIG. 1, and an inner surface **38**, as illustrated in FIG. 2. The outer surface **37** of the rear panel **30** will face outwardly and away from the athlete when the vest **10** is in use. Additionally, the inner surface **38** of the rear panel **30** will face toward the athlete when the vest **10** is in use. Upper rear rings **39A**, **39B** and **39C** are disposed at laterally spaced positions along the top edge **32** of the rear panel **30** and lower rear rings **39D**, **39E** and **39F** are disposed at laterally spaced positions along the bottom edge **34** of the rear panel **30**. The rings **39A-39F** may be mounted to swivel relative to the rear panel **30**.

[0039] The front panel **40** has opposite top and bottom edges **42** and **44** respectively and opposite first and second side edges **45** and **46** respectively. The front panel **40** also has an outer surface **47**, as illustrated in FIGS. 1 and 3, and a rear surface **48**, as illustrated in FIG. 2. The outer surface **47** of the front panel **40** will face outwardly and away from the athlete when the vest **10** is in use. Additionally, the inner surface **48** of the front panel **40** will face toward the athlete when the vest **10** is in use. Upper front rings **49A**, **49B** and **49C** are disposed at laterally spaced positions along the top edge **42** of the front panel **40** and lower front rings **49D**, **49E** and **49F** are disposed at laterally spaced positions along the bottom edge **44** of the front panel **40**. The rings **49A-49F** may be mounted to swivel relative to the front panel **40**.

[0040] First and second length-adjustable shoulder straps **50** and **52** extend between the top edge **32** of the rear panel **30** and the top edge **42** of the front panel **40**. The length of the first and second shoulder straps **50**, **52** can be extended or shortened to accommodate the size of the athlete. The first shoulder strap **50** is aligned substantially with the first side edges **35** and **45** of the rear panel **30** and the front panel **40** respectively. Similarly, the second shoulder strap **52** is aligned substantially with the second side edges **36** and **46** of the rear panel **30** and the front panel **40** respectively.

[0041] A first connecting panel **55** extends laterally from the first side edge **45** of the front panel **40** and a second connecting panel **56** extends laterally from the second side edge **46** of the front panel **40**. The first connecting panel **55** has an outer surface **57** as illustrated in FIG. 1, and a looped fabric **58** is provided on a region of the outer surface **57** of the first connecting panel **55** remote

from the front panel **40**. Similarly, the second connecting panel **56** has an outer surface **58**, and a looped fabric **59** is provided on a region of the outer surface **58** of the second connecting panel **56** remote from the front panel **40**.

[0042] A first securing belt **62** extends laterally from the first side edge **45** of the front panel **40** to a position beyond the first connecting panel **55**. FIG. **1** shows a folded end portion of the first securing belt **62** in broken lines and in a folded disposition so that the opposite surface of the end portion of the first securing belt **62** is visible. This end portion of the first securing belt **62** has a hooked fabric **63** with an array of hooks configured to releasably engage in the loops on the looped fabric **59** on the second connecting panel **56**. A belt loop **64** is attached to the first connecting panel **55** and the first securing belt **62** passes through the first belt loop **64** to hold the first securing belt **62** in proximity to the first connecting panel **55**, thereby reducing the likelihood of entanglement of the first securing belt **62** with other parts of the vest **10** and facilitating the secure snug attachment of the vest **10** to the athlete.

[0043] A second securing belt **72** extends laterally from the second side edge **46** of the front panel **40** to a position beyond the second connecting panel **56**. Again, FIG. **1** shows a folded end portion of the second securing belt **72** in broken lines and in a folded disposition so that the opposite surface of the end portion of the first securing belt **72** is visible. This end portion of the second securing belt **72** has a hooked fabric **73** with an array of hooks configured to releasably engage in the loops on the looped fabric **58** on the first connecting panel **55**. A belt loop **74** is attached to the second connecting panel **56** and the second securing belt **72** passes through the belt loop **74** to hold the second securing belt **62** in proximity to the second connecting panel **56**, thereby reducing the likelihood of entanglement of the second securing belt **72** with other parts of the vest **10** and facilitating the secure snug attachment of the vest **10** to the athlete.

[0044] The vest **10** can be used for training athletes in many sports and for many types of athletic maneuvers. One example relates to a batter in baseball. In this example, a baseball player wears the vest **10** in the manner shown in FIGS. **3** and **4** by placing his or her head into the space between the rear and front panels **30** and **40** and between the first and second shoulder straps **50** and **52**. Thus, the inner surface **38** of the rear panel **30** is opposed to the back of the baseball player and the inner surface **48** of the front panel **40** is adjacent the chest of the baseball player. The first and second connecting panels then are wrapped around the sides of the baseball player and at least partly overlap in the rear panel **30**. The first and second securing belts **62** and **72** then are pulled snugly toward the rear panel **30** and are positioned so that the hooked fabric **63** on the first securing belt **62** engages with the looped fabric **59** on the second connecting panel and so that the hooked fabric **73** on the second securing belt **72** engages the hooked fabric **58** on the first connecting panel **55**. Thus, the vest **10** is held securely, snugly and comfortably on the baseball player without impeding the ability of the player to move. More particularly, the athlete can move his or her arms freely in all directions and can twist his or her torso about a spine or other axis of rotation.

[0045] The vest **10** is used with at least one training coupler **80** having opposite first and second ends **82** and **84**, as shown in FIG. **5**. A clip **86** is attached to the first end **82** of the training coupler **80**. The clip **86** is configured to be attached securely but releasably to any one of the rings **39A-39F** in accordance with the training exercises to be carried out. In this example, the clip is attached by the coach or by the athlete, possibly in cooperation with the coach. The particular ring **39A-39F** to which the clip **86** is connected may define a pivot point of the upper body for the particular athletic maneuver that is the subject of the training exercise or may define an orbiting location that moves about a pivot point or pivot axis. The second end **84** of the elastic training coupler **80** may be held by the coach with a selected tension and alignment of the training coupler **80** relative to the back of the baseball player. More particularly, the coach will select one of the rings **39A-39F** to which the clip **86** on the first end **82** of the training coupler **80** will be attached. The selection of the particular ring **39A-39F** will be made in accordance with pivot point for the training exercise that is to be carried out. In a typical baseball batting training exercise, the coach holding the second end **84** of

the training coupler **80** will be facing the back of the batter and may be farther than the batter from the pitching mound or closer than the batter to the pitching mound in accordance with the particular training exercise. The training exercise will be carried out with the coach applying appropriate tension, for example, to prevent the batter from leaning forward and into the pitch or to prevent the batter from opening up too much during the course of a swing and to ensure that the baseball batter uses the proper pivot point about which force and acceleration will be developed. Additionally, a training exercise can enable the coach to provide a rearward force via the training coupler **80** on a shoulder that moves forward during a batting maneuver. This force will affect the actual or perceived stability sensed by the player and will cause the player to resist the force of the training coupler **80** by exerting forces opposite to the direction of force applied to or by the coupler **80**. Thus, the batter will exert additional forces toward the pitcher to avoid any instability caused by the forces exerted by the coupler. The muscle strength and muscle memory developed during such training exercises will cause the player to exert additional upper body force during a swing in an actual batting situation of a game. These exercises can be carried in some situations without an actual ball being thrown. Two or more training couplers **80** can be used simultaneously. For example first and second training couplers can be connected respectively to first and second rings. The rings to which the training couplers are attached may be two upper rings, two lower rings or an upper ring and a lower ring. Additionally, portions of each training coupler between the first and second ends may extend linearly or alternatively may be curved about the athlete to deliver a rotational or torsion force to the athlete as the second end of the training coupler **80** is pulled.

[0046] The vest **10** also can be used for training exercises of a pitcher. For example, the clip **86** at the first end of the training coupler **80** can be secured to one of the rings **39A-39F** of the vest **10** worn by a pitcher while the second end **84** of the training cord **80** may be secured to a rigid support. The pitcher may hold weight(s) on his or her shoulders with appropriate tension on the training cord. The weight may be a water-filled flexible plastic enclosure. The pitcher then can undergo the foot, leg and body movements carried out during a pitch, but without actually throwing a ball. The training coupler **80** will ensure proper positioning and alignment of the torso of the pitcher without impeding the body movements of the pitcher carried out during the course of a pitch and can define the most desirable and effective pivot point about which rotational forces can be developed. Additionally, for example, a training coupler **80** that is clipped to a rear shoulder area of the throwing arm will cause the pitcher to exert a forward and downward pivoting movement of the shoulder to counter the forces exerted by the training coupler **80**. These forces generated during a training session will develop muscle strength and establish trained pitching movements that will be carried over into an actual game situation.

[0047] The training vest **10** and the training coupler **80** can be used with one or more sensors **102**, as shown in FIG. 5. For example, FIG. 5 shows a sensor **102** attached to or incorporated into the training coupler **80**. The sensors **102** are illustrated schematically and generally and can take different forms. For example, sensors **102** can be used to measure the force that is being applied to the training coupler **80**, and hence the force that is being countered by the athlete. The sensor **102** also can be a strain sensor, a speed sensor, an accelerometer or the like. Output from the sensors **102** is transmitted, wirelessly in this embodiment, to a control unit **100** at a location remote from the athlete. The control unit **100** includes a memory that stores the measured data for subsequent presentation and analysis, for example, on a monitor. The illustrated embodiment also includes a camera device **104** that records images and transmits those images to the control unit **100** wirelessly. The control unit **100** can produce outputs that display body movements, force, speed and acceleration information at various stages in an athletic maneuver. Based on these data, the coach can decide to change the ring **39A-39F**, **49A-49F** to which the clip **86** at the second end **82** of the coupler **80** is attached to achieve a more favorable application of force by the athlete. Alternatively, based on these data, the coach can change the alignment of the force vector extending between the first and second ends **82**, **84** of the coupler **80**. The use of these electronic

and electro-optical components with the training vest **10** and training coupler **80** are illustrated only for a baseball batter. However, this combination can be used with pitching and with other sports. Additional sensors can be incorporated, particularly for other sports. For example, sensors can be incorporated into a punching bag, a golf tee, a lacrosse stick, a tennis racket, a baseball bat or the like. The vest **10** also can have indicia that can be sensed and traced by the camera device **104** during the course of the athletic maneuver

[0048] The invention has been described with respect to one embodiment. However, it is apparent that other variations of the training vest described and illustrated above fall within the scope of the invention. For example, the vest **10** and the training coupler **80** can be used in combination with a mat or template to show where feet should be positioned at the start and/or end of a batting or pitching movement. The vest **10** also can be used for training athletes in other sports, including, for example, hockey, lacrosse, boxing, shot putting, javelin throwing and others. Additionally, the training vest can be part of a sleeved or sleeveless garment that closely engages the torso, but permit free movement of the arms.

Claims

1. An athletic training assembly, comprising: a vest configured to be worn by an athlete, the vest including at least one attachment point; and at least one elongate training coupler having opposite first and second ends and an attachment mechanism at the first end of the at least one training coupler, the attachment mechanism being configured to influence a translational movement of a body part of the athlete or a rotational movement of a torso of the athlete about a pivot axis extending along the torso of the athlete.
2. The athletic training assembly of claim 1, wherein: the vest is formed from a flexible dimensionally stable material and includes opposite front and rear panels dimensioned respectively for fitting adjacent a chest and back of the athlete, left and right connecting panels extending between the front and rear panels, at least one of the left and right connecting panels being adjustable to enable secure fitting of the vest on the athlete, and left and right shoulder straps connecting the front panel to the rear panel.
3. The athletic training assembly of claim 2, wherein: the at least one attachment point comprises at least two front attachment points on the front panel and at least two rear attachment points on the rear panel.
4. The athletic training assembly of claim 1, wherein: the training coupler is dimensionally stable.
5. The athletic training assembly of claim 1, wherein: at least a part of the training coupler is elastic and flexible.
6. The athletic training assembly of claim 1, further comprising: at least one sensor attached to the vest or the training coupler.
7. The athletic training assembly of claim 6, wherein: the at least one sensor is configured for measuring at least one of: a position of the at least one sensor; forces applied at the at least one sensor; speed of the at least one sensor; and acceleration of the at least one sensor.
8. The athletic training assembly of claim 7, further comprising: a control unit communicating with the at least one sensor for receiving measured data from the at least one sensor, the control unit including a memory for storing the measuring data and an interface for outputting the measured data while performing an athletic movement.
9. The athletic training assembly of claim 8, further comprising: at least one camera for recording movement of the athlete while performing an athletic movement, the camera communicating with the control unit, and the control unit outputting a correlation of the measuring data received from the at least one sensor and images recorded by the camera.
10. A method for improving performance of an athletic maneuver that requires a pivoting movement of a part of an athlete's body about a pivot axis, the method comprising: securing a

garment around parts of the athlete's body that includes both the pivot axis and the part of the athlete's body that undergoes the pivoting movement about the pivot axis; providing a training coupler having opposite first and second ends, the first end having an attachment mechanism; attaching the attachment mechanism of the training coupler to an attachment location on the garment that is spaced from the pivot axis of the athlete's body and that will move with the parts of the athlete's body that undergo the pivoting movement about the pivot axis of the athlete's body; holding the training coupler at the second end of the training coupler so that an extending direction of at least a part of the training coupler from the first end to the second end is opposed to a moving direction of the attachment location during performance of the athletic maneuver; and applying tension between the first and second ends of the training coupler for biasing parts of the athlete's body adjacent to the attachment location in a biasing direction that is opposed to the moving direction of the attachment location during performance of the athletic maneuver.

11. The method of claim 10, wherein securing the garment around parts of the athlete's body that includes both the pivot axis and the part of the athlete's that undergoes the pivoting movement about the pivot axis comprises securing the garment around parts of the torso of the athlete's body.

12. The method of claim 11, wherein the attachment location is adjacent a shoulder of the athlete.

13. The method of claim 11, wherein: the attachment location on the garment is adjacent a part of the upper torso that will undergo a downward movement during performance of the athletic maneuver; and the holding of the coupler includes holding the coupler at a position higher than the attachment location at completion of the athletic maneuver.

14. The method of claim 10, wherein: holding the training coupler at a location spaced from the first end of the training coupler includes manually holding the training coupler while maintaining tension on the training coupler.

15. The method of claim 10, wherein: the training coupler is formed from an elastic material and holding the training coupler at the second end of the training coupler includes attaching the second end of the training coupler releasably to a rigid support.

16. A method for training an athlete to perform an athletic maneuver, the method comprising: fitting a dimensionally stable garment on the athlete; applying at least one external force to at least one force application location on the dimensionally stable garment, the at least one force application location being at least one location that will move in a moving direction during performance of the athletic maneuver, and the at least one external force is applied in at least one direction that is different from the at least one moving direction of the force application location during performance of the athletic maneuver; and performing the athletic maneuver while overcoming the at least one external force, thereby training the athlete to vary directions and magnitudes of forces generated to perform the athletic maneuver.

17. The method of claim 16, wherein the at least one external force is at least one of a linear force and a rotational force.

18. The method of claim 16, wherein the dimensionally stable garment is dimensioned and configured to exert a compressive force on the athlete.

19. The method of claim 16, further comprising: providing at least one elongated training coupler having opposite first and second ends; removably coupling the first end of the at least one elongated training coupler to the force application location on the dimensionally stable garment; and applying the at least one external force to the at least one force application location on the dimensionally stable garment comprises applying a pulling force on the second end of the at least one elongated training coupler.
